**Industrial uses of alcoholic fermentation:**

1. In bakeries, it is used for preparing bread, cakes, biscuits.
2. In beverage industries for preparing wine and alcoholic drinks.
3. In producing vinegar and in tanning, curing of leather.
4. Ethanol is used to make gasohol (a fuel that is used for cars in Brazil).

**2. Lactic acid fermentation**

Some bacteria (*Bacillus*), fungi and muscles of vertebrates produce lactic acid from pyruvic acid (Table 14.3).

$$2CH_3COCOOH + 2NADH + 2H^+ \xrightarrow[\text{Lactate dehydrogenase}]{} 2CH_3CHOHCOOH + 2NAD^+$$

Pyruvic acid → Lactic acid

**3. Mixed acid fermentation**

This type of fermentation is a characteristic feature of Enterobacteriaceae and results in the formation of lactic acid, ethanol, formic acid and gases like $CO_2$ and $H_2$.

**Characteristics of Anaerobic Respiration**

1. Anaerobic respiration is less efficient than the aerobic respiration (Figure 14.12).
2. Limited number of ATP molecules is generated per glucose molecule (Table 14.4).
3. It is characterized by the production of $CO_2$ and it is used for Carbon fixation in photosynthesis.

**Table 14.4: Net products from one molecule of Glucose under Glycolysis and Anaerobic respiration.**

| Stage | Substrate level ATP production | Reduced $NAD^+$ | Total ATP |
|---|---|---|---|
| Glycolysis | 2 | 2* | 8 |
| Anaerobic respiration | 2 | 2 reduced $NAD^+$ re-oxidised | 2 |

*One reduced $NAD^+$ equivalent to 3 ATPs

**Demonstration of alcoholic fermentation**

Take a Kuhne's fermentation tube which consists of an upright glass tube with side bulb. Pour 10% sugar solution mixed with baker's yeast into the fermentation tube the side tube is filled plug the mouth with lid. After some time, the glucose solution will be fermented. The solution will give out an alcoholic smell and level of solution in glass column will fall due to the accumulation of $CO_2$ gas. It is due to the presence of zymase enzyme in yeast which converts the glucose solution into alcohol and $CO_2$. Now introduce a pellet of KOH into the tube, the KOH will absorb $CO_2$ and the level of solution will rise in upright tube (Figure 14.13).

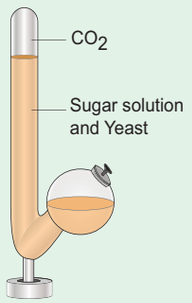



**Figure 14.13:** Kuhne's fermentation experiment

**Activity**

Take a bottle filled with warm water mixed with baker's yeast and sugar. After some time, you will notice water bubbling as yeast produces carbon dioxide. Attach a balloon to the mouth of the bottle. After 30 minutes you'll notice balloon standing upright (Figure 14.14).

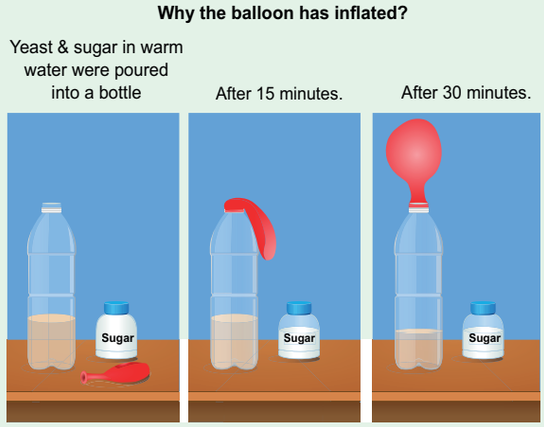



**Figure:** 14.14: Air balloon activity



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